Gengcong Yang

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Education

University of Texas at Austin 08/2023 – 12/2024
*Master in **Computer Science**; GPA: 3.7 / 4.0* Austin, United States

Tsinghua University 08/2019 – 06/2022
*Master in **Control Engineering**, Department of Automation; GPA: 3.6 / 4.0* Beijing, China
Thesis: Research on Scene Graph Generation via Uncertainty Modeling; Advisor: Yujiu Yang

Sun Yat-sen University 08/2015 – 06/2019
*Bachelor in **Software Engineering**; GPA: 3.8 / 4.0* Guangzhou, China
Thesis: Research on Diverse and Distinctive Image Captioning Based on Reinforcement Learning

National Chiao Tung University 09/2017 – 01/2018
*Exchange Student in **Computer Science**; GPA: 3.8 / 4.0* Hsinchu, Taiwan

Publications

Diamond: Harnessing GPU Resources for Scientific Deep Learning. *IEEE eScience*, 2025.

Haotian Xie, Rohan Marwaha, Minu Mathew, Song Bian, **Gengcong Yang**, et al.

- Contributed to the design of a distributed job-submission pipeline enabling scalable, multi-node GPU training on a platform for the scientific machine learning lifecycle; funded by **NSF**.

Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation. *CVPR*, 2021.

Gengcong Yang, Jingyi Zhang, Yong Zhang, Baoyuan Wu, Yujiu Yang.

- Proposed a probabilistic framework and designed a plug-and-play uncertainty modeling module to capture semantic ambiguity and multiple plausible visual relationships in scene graphs.

Research Experience

Texas Advanced Computing Center, *Graduate Research Assistant* Austin, 01/2024 – 12/2024

- Supervisor: Yinzhi Wang
- Contributed to the early-stage development of Diamond, an NSF-funded, scalable platform supporting the full machine learning research lifecycle on HPC resources.
- Designed and implemented a distributed job-submission pipeline enabling users to containerize environments and train ML models across multiple GPU nodes.

KAUST, *Research Intern* Remote, 02/2021 – 07/2021

- Supervisor: Bernard Ghanem
- Implemented a differentiable RoI Pooling method with CUDA and C++ for the temporal action detection task on videos, achieving an experimental improvement of 6%.

Tencent AI Lab, *Computer Vision Researcher Intern* Shenzhen, 07/2019 – 07/2020

- Supervisor: Baoyuan Wu
- Developed a novel modeling method for polysemous visual-relationship representation and applied for a technology patent.
- Contributed to the design and experimentation of the probabilistic uncertainty modeling framework later published at CVPR 2021.

Industry Experience

TikTok USDS, Emerging Product, *Machine Learning Engineer* San Jose, 07/2026 – Present

- Led safety guardrails to detect and block unsafe prompts and generated videos for Seedance, a video generation model.
- Built and scaled GPU model-serving infrastructure for U.S. deployment.
- Developed LLM-based agents that automate the team's operational workflows, reducing manual effort.

Uber, Driver Offer Pricing Team, *Machine Learning Engineer* Sunnyvale, 01/2025 – 07/2026

- Developed driver fare prediction models used as core pricing signals within Uber's marketplace pricing system.
- Improved observability and reliability infrastructure for pricing prediction pipelines running at scale.

ByteDance, E-commerce, *Machine Learning Engineer* Shanghai, 06/2022 – 06/2023

- Developed image search models and a retrieval-ranking framework for the TikTok Shop similar-product search project.
- Constructed large-scale datasets for post-pretraining and quantized object detection models to improve serving efficiency.

Tencent, *Machine Learning Engineering Intern* Beijing, 06/2021 – 08/2021

- Focused on the retrieval stage of the recommendation system for an early-stage short-video platform.
- Built an offline evaluation pipeline for retrieval models and experimented with pairwise training for a two-tower model.

Patents

Visual Relationship Recognition Method Based on Artificial Intelligence

- Inventors: **Gengcong Yang**, Yong Zhang, Baoyuan Wu, Yanbo Fan, Zhifeng Li, Yujiu Yang, Wei Liu
- Patent Number: CN202011108484
- Granted Date: 03/05/2024

Selected Awards

Tsinghua SIGS Professional Practice Scholarship	2022
Tsinghua Excellent Scholarship First Prize	2021
SYSU Outstanding Undergraduate Thesis Award	2019
SYSU Excellent Scholarship Second Prize	2018
National Encouragement Scholarship	2017, 2018
SYSU Xian Weijian Overseas Study Funding	2017
Interdisciplinary Contest In Modeling Honorable Mention	2017
SYSU Excellent Scholarship Third Prize	2016, 2017